



CHECKLIST

SMALL CRAFT - LIQUID-FUELLED GALLEY STOVES AND HEATING APPLIANCES

Ref.: EN ISO 14895:2016 (ISO 14895:2016)

Checklist_Evaluation_Module B_G en260703

Watercraft manufacturer:	
Watercraft model name:	



Subject to check	Clause	Requirements	Checked ?
1 Is petrol used as fuel or for priming of the appliances?	4.2	[No ?]	
2 Heaters and their exhaust systems are installed outside areas flammable vapours can accumulate.	4.3	[Yes / NA ?]	
3 Marine environment has been considered for design and installation of appliances (e.g. vibration, craft movement, humidity, corrosion).	4.4	[Yes ?]	
4 Appliance control is readily accessible and located to minimize possible injury from burners and/or hot components where being used.	4.5	[Yes ?]	
5 Outgoing combustion of heaters pass through sealed ductwork terminating outside of the craft.	4.6	[Yes / NA ?]	
6 Where user awareness for the safe operation of an appliance is required, a durable, permanently legible sign covering the operation, including the refuelling procedure if applicable, and any unique hazards involved with its use, shall be provided on or in the immediate vicinity as per clause 8.	4.7	[Yes / NA ?]	
7 Appliances requiring priming shall be fitted with a readily accessible drip pan to contain any fuel overflowing from the priming fuel container under conditions of pitch and heel as applicable (see 6.8 for stoves and 7.1 for heaters).	4.9	[Yes / NA ?]	
8 The drip pan shall be at least 20 mm deep.	4.9	[Yes / NA ?]	
9 Appliances shall not feature open flame pilot lights. If pilot lights are integral then they shall be installed so that outgoing products of combustion pass through sealed ductwork terminating outside the craft.	4.10	[Yes / NA ?]	
10 The appliance and any associated remote fuel tank is securely fastened to the craft.	5.1	[Yes / NA ?]	
11 The risk of injury or damage is minimized by the position of the appliance.	5.2	[Yes ?]	
12 The heater and/or heater exhaust components shall not constitute a risk of fire, even in the case of overheating.	5.3	[Yes / NA ?]	
13 For stoves with non-integral fuel tank: a readily accessible shut-off valve, non-integral with the stove is located near the fuel tank.	5.4	[Yes / NA ?]	
14 The valve shall close against the flow of fuel.	5.4	[Yes / NA ?]	
15 The on- and off-position and closing direction of the valve shall be indicated.	5.4	[Yes / NA ?]	

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16	If the fuel tank is located outside of the galley and mounted higher than the stove or back siphoning is not prevented, a second valve is fitted ready accessible and reachable outside of zone II as per ISO 9094.	5.4	[Yes / NA ?]	
17	Electrical operated shut-off devices, which are interrupting the fuel flow when not activated, are permitted.	5.4	[Yes / NA ?]	
18	For heaters with a non-integral tank: a fuel shut-off device shall be located near the remote fuel tank.	5.4	[Yes / NA ?]	
19	The valve shall close against the flow of fuel.	5.4	[Yes / NA ?]	
20	The on- and off-position and closing direction of the valve shall be indicated.	5.4	[Yes / NA ?]	
21	Electrical operated shut-off devices, which are interrupting the fuel flow when not activated, are permitted.	5.4	[Yes / NA ?]	
22	Any remote fuel tank is installed outside Zone II as per ISO 9094.	5.5	[Yes / NA ?]	
23	Fuel lines are solid metallic piping or flexible hoses meeting ISO 8469, clause 5 outside and ISO 7840 inside of engine rooms.	5.7	[Yes / NA ?]	
24	Joints in the distribution pipes or hoses are kept to a minimum.	5.8	[Yes / NA ?]	
25	Remote tanks filling point shall be outside of Zone II as per ISO 9094.	5.9	[Yes / NA ?]	
26	The combustion air inlet is positioned or guarded that it cannot be blocked.	5.10	[Yes / NA ?]	
27	Heating air shall be fresh air or re-circulated and drawn from clean area not likely to be contaminated by exhaust fumes.	5.11	[Yes / NA ?]	
28	Any ductwork is securely fastened.	5.11	[Yes / NA ?]	
29	If the appliance consumes air for combustion from habitable spaces and no other means of ventilation is provided to the habitable space, then fixed ventilation shall be provided of size equal to or greater than the appliance's combustion air intake.	5.12	[Yes / NA ?]	
30	Exhaust outlet shall be located to avoid emissions from entering the habitable space(s) of the craft.	5.13	[Yes / NA ?]	
31	Any ducting used to route the hot heating air through the vessel shall be so positioned or protected that no injury or damage could be caused if it were to be touched.	5.14	[Yes / NA ?]	
32	Any brackets or ties used to support or secure such parts as per 5.15 and/or 5.16 shall be of a suitable heat resistant material. If the heater is installed in a locker, then suitable steps shall be taken to protect any surrounding material and contents from heat damage.	5.17	[Yes / NA ?]	
33	The duplicate label (see Clause 8.1) shall be affixed where it is clearly visible if the original appliance label is obscured.	5.18	[Yes / NA ?]	
34	If a portable tank is used, it shall be of suitable design and labelled for the type of fuel used and shall have provisions to be secured on-board the craft.	5.19	[Yes / NA ?]	
35	Stove design and construction.	6		
36	Liquid-fuel priming pans or troughs shall be secured to the burner or heat generator so that their relationship is maintained.	6.1	[Yes / NA ?]	
37	Priming pans or troughs shall be designed to contain fuel without spillage under conditions of pitch or roll of the craft to 15° in any direction.	6.2	[Yes / NA ?]	

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38	Oven doors shall be provided with a means to prevent unintentional opening due to force from sliding food and utensils.	6.9	[Yes / NA ?]	
39	Operating, maintenance and installation instructions shall be supplied with every stove, drawing particular attention to set up, maintenance, regular operation, prevention of risks and risk management.	6.10	[Yes / NA ?]	
40	Heater design and construction.	7		
41	A clearly visible tell-tale shall be provided and shall indicate when the combustion heater is switched on or off.	7.7	[Yes / NA ?]	
42	Operating, maintenance and installation instructions shall be supplied with every heater, drawing particular attention to set up, maintenance, regular operation, prevention of risks and risk management.	7.8	[Yes / NA ?]	
43	Marking	8		
44	Each appliance shall be permanently marked or labelled with the following information. A duplicate label shall be provided by the manufacturer:	8.1		
45	- manufacturer's name or trademark;	8.1	[Yes ?]	
46	- model number;	8.1	[Yes ?]	
47	- rated heating output (for heaters);	8.1	[Yes ?]	
48	- serial number;	8.1	[Yes ?]	
49	- fuel type(s) on integral tanks at the opening used for filling or heater ID plate;	8.1	[Yes ?]	
50	- operating voltage (where relevant).	8.1	[Yes ?]	
51	Label: Open flame stoves label, in language acceptable in the country of use.  WARNING: OPEN FLAME STOVES PRODUCE CARBON MONOXIDE. AVOID ASPHYXIATION. MAINTAIN OPEN VENTILATION WHEN STOVE IS IN USE Affixed: On or adjacent to open flame stoves.	8.2	[Yes / NA ?]	
52	Label: Stoves with integral fuel tanks label in language acceptable in the country of use.  WARNING: POSSIBLE HAZARD OF EXPLOSION AND FIRE. TURN OFF STOVE BURNERS BEFORE FILLING FUEL CONTAINER Affixed: On, or adjacent to, the stove.	8.3	[Yes / NA ?]	

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- 53 **Label:** Non-pressurized stoves with integral tanks designed to have the fuel container removed for filling label in language acceptable in the country of use. 8.4 [Yes / NA ?]



WARNING: POSSIBLE HAZARD OF EXPLOSION AND FIRE. REMOVE FUEL CONTAINER FROM STOVE BEFORE FILLING. TURN OFF STOVE BURNERS BEFORE FILLING. FILL FUEL CONTAINER AWAY FROM STOVE

Affixed: On, or adjacent to, the stove.

- 54 Openings for filling fuel tanks shall be identified to indicate the type of fuel to be used with. 8.5 [Yes / NA ?]
NOTE: The word "FUEL" alone shall not be used.

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

Subject to check	Clause	Requirements	Checked ?
55 Appliance is installed according to manufacturer's instructions.	4.1	[Yes ?]	
56 Protection shall be provided to prevent human contact with exposed parts of the heating system exceeding a surface temperature of 110 °C.	5.15	[Yes ?]	
57 The heater or the heated medium shall not be liable to cause burns to persons. The surface temperature of any part of the heating system likely to come into contact with any person during normal craft operation shall not exceed a temperature of 85 °C.	5.16	[Yes / NA ?]	
58 Pressurized liquid-fuel tanks integral with a stove shall be equipped with a pressure relief valve designed to release at not more than twice the vapour pressure of the fuel used at 60 °C.	6.3	[Yes / NA ?]	
59 Pressurized liquid-fuel tanks integral with a stove shall be shielded or insulated so that, under continuous operation at maximum heat, the pressure in the tank will not exceed 50 % of the relief valve setting.	6.4	[Yes / NA ?]	
60 Pressurized liquid-fuel tanks integral with a stove shall be designed to withstand four times the relief-valve setting.	6.5	[Yes / NA ?]	
61 Pressurized liquid-fuel tanks integral with a stove shall be tested to withstand a minimum internal pressure of two times the design working pressure or 700 kPa, whichever is greater.	6.6	[Yes / NA ?]	
62 Stoves shall be capable of operating during periods of craft pitch or heel at angles up to 15° in any direction sustained at the maximum angle for at least 15 s. Stoves in monohull sailing craft shall be capable of operation at sustained angles of heel of 30°. This may be obtained by the use of gimbals.	6.7	[Yes / NA ?]	

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63	Means shall be provided on or adjacent to stove-top cooking surfaces to prevent both deep and shallow cooking utensils from sliding across or off the stove, at pitch angles of up to 15° for all craft, and heel angles of 15° for engine-driven craft and sailing multihull and 30° for monohull sailing craft. This may be obtained by gimbals.	6.8	[Yes / NA ?]	
64	Heaters shall be designed and constructed to meet the following general requirements: - be suitable for marine use; - be able to operate at least of 15° heel or pitch in any direction; - have overheat control devices.	7.1	[Yes / NA ?]	
65	Heaters shall not pollute the heating air. The combustion circuit of the heat exchanger shall be subjected to a leakage test to ensure that exhaust gasses cannot enter the heated air intended for the habitable space.	7.2	[Yes / NA ?]	
66	The temperature of the heated air entering the habitable space shall not exceed 150 °C to be measured at the centre of the heating air outlet.	7.3	[Yes / NA ?]	
67	An integral flame failure device shall be installed in every heater. This device shall recognize a flame failure and shut the heater down in a controlled manner.	7.4	[Yes / NA ?]	
68	In the event of a failed start the heater shall be designed to avoid any fuel overflowing. This may be achieved by supplying a safety lock out system following a pre-determined number of failed start attempts or by supplying of return lines.	7.5	[Yes / NA ?]	
69	If a combustion air blower is fitted, a delayed shut-off shall be provided, even in the event of overheating or in the event of an interrupted fuel supply.	7.6	[Yes / NA ?]	

Instructions/Warnings to be included in the owner's manual

70	Stoves:			
71	Instructions to read the stove's owner's manual.	Annex A	[Yes / NA ?]	
72	Instructions for priming, lighting and use of the stove, also for turning off the stove and fuel supply.	Annex A	[Yes / NA ?]	
73	Type of fuel used (such as alcohol type, diesel, kerosene) and cautions and possible hazards with other fuels that are not suitable for use.	Annex A	[Yes / NA ?]	
74	Instructions for the refuelling procedure, including possible hazard warnings concerning fire and explosion.	Annex A	[Yes / NA ?]	
75	Cleaning and maintenance of the stove and fuel system.	Annex A	[Yes / NA ?]	

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X

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76	Warnings and cautions with at least the following informational elements: - open-flame stoves consume oxygen and to draw the owner's attention to the warning label required in 8.2; - insufficient oxygen supply can cause asphyxiation; - maintain ventilation open when stove is in use; - do not use the open-flame stove for space or comfort heating; - stove exhaust components can be hot during and directly after operation; - ensure stove inlets and exhaust outlets are not obstructed while stove is in operation; - turn stove off during refuelling.		
77	Recommendations for storage of spare fuel: -keep spare fuel to a minimum and in containers of corrosion-resistant material; -keep stowed outside the accommodation spaces; -keep stowed in areas where the temperature cannot exceed 60 °C.	Annex A	[Yes / NA ?]
78	Heaters:	Annex B	
79	Instructions to read the heater's owner's manual.	Annex B	[Yes / NA ?]
80	Instructions for use of the heater, also for turning off and fuel supply. These instructions shall include ensuring heater's cool down cycle is not interrupted and turning off heater fuel supply if manual valve is fitted.	Annex B	[Yes / NA ?]
81	Type of fuel used (such as alcohol type, diesel, kerosene) and cautions and possible hazards with other fuels that are not suitable for use.	Annex B	[Yes / NA ?]
82	Instructions for the refuelling procedure, including possible hazard warnings concerning fire and explosion.	Annex B	[Yes / NA ?]
83	Specify inspection and/or maintenance intervals.	Annex B	[Yes / NA ?]
84	Recommendations for storage of spare fuel: -keep spare fuel to a minimum and in containers of corrosion-resistant material; -keep stowed outside the accommodation spaces; -keep stowed in areas where the temperature cannot exceed 60 °C.	Annex B	[Yes / NA ?]
85	Recommendations for storage of spare fuel: - keep spare fuel to a minimum and in containers of corrosion-resistant material; - keep stowed outside the accommodation spaces; - keep stowed in areas where the temperature cannot exceed 60 °C.	Annex B	[Yes / NA ?]

Comments:

compliant: Yes or V

not applicable: NA

follow up on variation report: Rpt or X